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JavaScript Object Protection

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Object Protection Methods

```
// Prevents re-assignment
const car = {type:"Fiat", model:"500", color:"white"};

// Prevents adding object properties
Object.preventExtensions(object)

// Returns true if properties can be added to an object
Object.isExtensible(object)

// Prevents adding and deleting object properties
Object.seal(object)

// Returns true if object is sealed
Object.isSealed(object)

// Prevents any changes to an object
Object.freeze(object)

// Returns true if object is frozen
Object.isFrozen(object)
```

Using const

The most common way to protect an object from being changed is by using the `const` keyword.

With `const` you can not re-assign the object, but you can still change the value of a property, delete a property or create a new property.

The `Object.preventExtensions()` method prevents adding properties to an object.

Example

```
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Prevent Extensions
Object.preventExtensions(person);

// This will throw an error
person.nationality = "English";
```

Try it Yourself »

Since arrays are objects, arrays can be prevented from extensions too:

Example

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.preventExtensions(fruits);

// This will throw an error:
fruits.push("Kiwi");
```

Try it Yourself »

JavaScript Object.isExtensible()

You can use `Object.isExtensible()` to check if an object is extensible.

The `Object.isExtensible()` returns true if an object is extensible.

Examples

```
// Prevent Extensions
Object.preventExtensions(person);

// This will return false
let answer = Object.isExtensible(person);
```

Try it Yourself »

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Prevent Extensions
Object.preventExtensions(fruits);

// This will return false
let answer = Object.isExtensible(fruits);
```

Try it Yourself »

JavaScript Object.seal()

The `Object.seal()` method prevents additions or deletions of new properties.

The `Object.seal()` method makes existing properties non-configurable.

The `Object.isSealed()` method can be used to check if an object is sealed.

Note

The `Object.seal()` method will fail silently in non-strict mode and throw a TypeError in strict mode.

Example

```
"use strict"
// Create Object
const person = {
  firstName: "John",
  lastName: "Doe",
```

```
// Seal Object
Object.seal(person)

// This will throw an error
delete person.age;
```

Try it Yourself »

Since arrays are objects, arrays can be sealed too:

Example

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.seal(fruits);

// This will throw an error:
fruits.push("Kiwi");
```

Try it Yourself »

JavaScript Object.isSealed()

The `Object.isSealed()` method can be used to check if an object is sealed.

The `Object.isSealed()` returns true if an object is sealed.

Examples

```
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Seal Object
Object.seal(person);

// This will return true
let answer = Object.isSealed(person);
```

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Seal Array
Object.seal(fruits);

// This will return true
let answer = Object.isSealed(fruits);
```

Try it Yourself »

JavaScript Object.freeze()

The `Object.freeze()` method prevents any changes to an object.

Frozen objects are read-only.

No modification, addition or deletion of properties are allowed.

Note

The `Object.freeze()` method will fail silently in non-strict mode and throw a TypeError in strict mode.

Example

```
"use strict"
// Create Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Freeze Object
Object.freeze(person)
```

Since arrays are objects, arrays can be frozen too:

Example

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.freeze(fruits);

// This will trow an error:
fruits.push("Kiwi");
```

Try it Yourself »

JavaScript Object.isFrozen()

The `Object.isFrozen()` method can be used to check if an object is frozen.

The `Object.isFrozen()` returns true if an object is frozen.

Examples

```
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Freeze Object
Object.freeze(person);

// This will return true
let answer = Object.isFrozen(person);
```

Try it Yourself »

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.freeze(fruits);
```

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